

PERIODIC BROADCASTS OF DENSITY $1/10$, $1/8$ AND $5/36$: CONFIRMING AND STRENGTHENING THREE CONJECTURED DENSITY BOUNDS OF CERVANTES AND HARRIS

ABSTRACT. Cervantes and Harris conjectured that the optimal (t, r) broadcast densities of the infinite truncated square tiling graph $H_{\infty, \infty}$ satisfy $\delta_{4,2} \leq 9/80$, $\delta_{4,3} \leq 1/7$ and $\delta_{4,4} \leq 1/6$, leaving all three as conjectures short of a detailed proof; their witnesses appear there as figures. We exhibit three explicit lattice-periodic broadcasts, each written as a congruence in the vertex coordinates, of densities $1/10 < 9/80$, $1/8 < 1/7$ and $5/36 < 1/6$. So all three conjectured inequalities hold, and all three conjectured upper bounds can be improved. A capping argument on the published coordination sequence $1, 3, 5, 8$ gives $\delta_{4,2} \geq 1/13$, so $1/13 \leq \delta_{4,2}(H_{\infty, \infty}) \leq 1/10$; we determine none of the three optimal densities. We note the honest baseline: since every $(3, 1)$ broadcast is a $(4, 2)$ broadcast, $\delta_{4,2} \leq 1/8$ was already implied by a theorem of the same paper, so at $r = 2$ the gain over the bound the source itself implies is from $1/8$ to $1/10$. All exhibited objects are finite and every claim about them is checkable by hand from the tables printed here.

1. THE CONJECTURE AND WHAT IS PROVED

Let G be a graph, $t \geq 2$ and $r \geq 1$ integers. Following Blessing, Johnson, Mauretour and Insko [1] in the formulation quoted by Cervantes and Harris [2], a set $T \subseteq V(G)$ is a (t, r) *broadcast* if every $u \in V(G)$ satisfies

$$(1) \quad f(u) := \sum_{\substack{v \in T \\ d(u,v) < t}} (t - d(u, v)) \geq r.$$

The distance cutoff in (1) is strict, so at $t = 4$ a tower contributes $4, 3, 2, 1$ to the vertices at distance $0, 1, 2, 3$ from it and nothing at distance ≥ 4 . The *density* of T is $\lim_{n \rightarrow \infty} |T \cap V(H_n)|/|V(H_n)|$ along an exhausting sequence of finite subgraphs, and $\delta_{t,r}(G)$ is the infimum of the density over all (t, r) broadcasts; Cervantes and Harris take this definition from [3].

Let $H_{\infty, \infty}$ denote the infinite *truncated square tiling graph*, the 3-regular Archimedean 4.8.8 net. Cervantes and Harris determine $\delta_{t,r}$ for $(t, r) \in \{(2, 1), (2, 2), (3, 1), (3, 2), (3, 3), (4, 1)\}$ and then conjecture [2]:

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The optimal densities of $H_{\infty,\infty}$ satisfy

$$\delta_{4,2}(H_{\infty,\infty}) \leq \frac{9}{80}, \quad \delta_{4,3}(H_{\infty,\infty}) \leq \frac{1}{7}, \quad \text{and} \quad \delta_{4,4}(H_{\infty,\infty}) \leq \frac{1}{6}.$$

The conjecture is a bound in each of its three coordinates, not an equality. Its authors write that they “provide broadcasts that achieve the bounds” in three figures, but that “given the many cases needed, and short of a detailed proof, we leave the statements as conjectures” [2]; the accompanying problem of that paper asks for a proof that the sets of *those figures* dominate $H_{\infty,\infty}$ with the stated densities. Because $\delta_{t,r}$ is an infimum, each inequality is an existence statement, so a single explicit periodic broadcast settles it, and a broadcast of smaller density improves it.

Theorem 1.

$\delta_{4,2}(H_{\infty,\infty}) \leq \frac{1}{10} < \frac{9}{80}$, $\delta_{4,3}(H_{\infty,\infty}) \leq \frac{1}{8} < \frac{1}{7}$, $\delta_{4,4}(H_{\infty,\infty}) \leq \frac{5}{36} < \frac{1}{6}$, witnessed by the lattice-periodic broadcasts T_5, T_6, T_7 of §3. In particular all three conjectured inequalities hold, and each of the three conjectured upper bounds can be improved.

Theorem 2. $\delta_{4,r}(H_{\infty,\infty}) \geq r/C(r)$ where $C(r) = \min(4, r) + 3 \min(3, r) + 5 \min(2, r) + 8 \min(1, r)$. In particular $\delta_{4,2} \geq 2/26 = 1/13$, $\delta_{4,3} \geq 1/10$ and $\delta_{4,4} \geq 4/31$.

Combining Theorems 1 and 2,

$$\frac{1}{13} = 0.076923\dots \leq \delta_{4,2}(H_{\infty,\infty}) \leq \frac{1}{10} = 0.1,$$

with the conjectured $9/80 = 0.1125$ lying strictly above the upper end. We determine none of the three optimal densities.

We state one baseline explicitly, because it is easy to overlook.

Proposition 3. Every $(3, 1)$ broadcast of a graph is a $(4, 2)$ broadcast, so $\delta_{4,2} \leq \delta_{3,1}$. Consequently $\delta_{4,2}(H_{\infty,\infty}) \leq 1/8$ already follows from the theorem of [2] on $\delta_{3,1}$, which gives $\delta_{3,1}(H_{\infty,\infty}) \leq 1/8$.

Proof. Let T be a $(3, 1)$ broadcast and u a vertex. Then $f_3(u) \geq 1$, so some $v \in T$ has $3 - d(u, v) \geq 1$ with $d(u, v) < 3$, hence $d(u, v) \leq 2$. At $t = 4$ that one tower alone contributes $4 - d(u, v) \geq 2$. $\square$

So for $r = 2$ the honest reading of Theorem 1 is a chain $1/8 \rightarrow 9/80 \rightarrow 1/10$: the conjectured bound already strengthens the one implied by the source paper, and $1/10$ strengthens it further. The same argument applied to the source’s theorems on $\delta_{3,2}$ and $\delta_{3,3}$ gives $\delta_{4,3} \leq \delta_{3,2} \leq 1/6$ and $\delta_{4,4} \leq \delta_{3,3} \leq 1/4$, both weaker than the corresponding conjectured values.

2. COORDINATES, SHELL SIZES, AND THE DENSITY OF A PERIODIC SET

We use the normal form of [2]: with the origin fixed, every vertex of $H_{\infty,\infty}$ is a pair $(a, (x, y))$ with $a \in \mathbb{Z}_4$ and $(x, y) \in \mathbb{Z}^2$, where (x, y) indexes the squares of the underlying square tiling and a numbers the four vertices of

that square clockwise from the top. We write $a@(\mathbf{x}, \mathbf{y})$. Adjacency is given by six rules: the four sides of each square,

$$0@(\mathbf{x}, \mathbf{y}) \sim 1@(\mathbf{x}, \mathbf{y}), \quad 1@(\mathbf{x}, \mathbf{y}) \sim 2@(\mathbf{x}, \mathbf{y}),$$

$$2@(\mathbf{x}, \mathbf{y}) \sim 3@(\mathbf{x}, \mathbf{y}), \quad 3@(\mathbf{x}, \mathbf{y}) \sim 0@(\mathbf{x}, \mathbf{y}),$$

and the two edge stubs that join neighbouring squares,

$$0@(\mathbf{x}, \mathbf{y}) \sim 2@(\mathbf{x}, \mathbf{y} + \mathbf{1}), \quad 1@(\mathbf{x}, \mathbf{y}) \sim 3@(\mathbf{x} + \mathbf{1}, \mathbf{y}).$$

The graph is 3-regular and the translation action of $\mathbb{Z}^2$, $a@(\mathbf{x}, \mathbf{y}) \mapsto a@(\mathbf{x} + \mathbf{p}, \mathbf{y} + \mathbf{q})$, is by automorphisms.

The vertex figure of the truncated square tiling — one square and two octagons, with no cycle of length 3, 5, 6 or 7 through a vertex — is a local condition, and the accompanying program confirms it at 36 sampled vertices of the presentation above; we do not verify that no other Archimedean net has that vertex figure.

Only one numerical fact about $H_{\infty, \infty}$ is needed below. From each of the four vertex types the numbers of vertices at distance 0, 1, 2, 3, 4 are

$$(2) \quad 1, 3, 5, 8, 11,$$

so $|B_2(v)| = 9$ and $|B_3(v)| = 17$ for every v . These are the source's own integers: the coordination sequence 3, 5, 8, 11, ... is quoted there from [7], and the count $|P_v(4)| = 1 + 3 + 5 + 8 = 17$ appears verbatim in a figure caption of [2].

Finally, the device that makes every claim below finite. Let $L \subseteq \mathbb{Z}^2$ be a sublattice of index s and let T be a set of vertices invariant under L . Then T meets a fundamental domain of $4s$ vertices in some number k of them, and since the boundary of a growing window is $O(\text{perimeter})$,

$$(3) \quad \text{density}(T) = \frac{k}{4s} \quad \text{exactly, with no boundary correction,}$$

while f is constant on each of the $4s$ classes because L acts by automorphisms preserving T . So verifying that an L -periodic T is a (t, r) broadcast is a finite check: evaluate (1) once per class, reading membership of T from its defining congruence. Each of the three broadcasts below is presented that way, with its full table of f -values, so no computation is left to the reader.

3. THE THREE WITNESSES

Throughout, $L = \langle (A, 0), (B, D) \rangle$ has index AD and we use the class invariants stated with each pattern; invariance of the invariant under both generators is immediate and is checked in each case. The labelling T_5, T_6, T_7 is that of the accompanying program, which also checks patterns not needed here.

T_5 : $\delta_{4,2} \leq 1/10$. Let $L_5 = \langle (5,0), (4,1) \rangle$, of index 5, so a fundamental domain has 20 vertices, and put $i(x, y) = (x+y) \bmod 5$, which is L_5 -invariant since both generators shift $x+y$ by $5 \equiv 0$. Set

$$T_5 = \{ 0@ (x, y) : x+y \equiv 0 \pmod{5} \} \cup \{ 2@ (x, y) : x+y \equiv 3 \pmod{5} \}.$$

This is 2 of 20 classes, so the density is exactly $2/20 = 1/10$, against the conjectured $9/80$. The complete table of f , with $u = a@(i, 0)$:

	$i = 0$	$i = 1$	$i = 2$	$i = 3$	$i = 4$
$a = 0$	4*	2	4	2	3
$a = 1$	4	3	4	3	2
$a = 2$	2	4	2	4*	3
$a = 3$	3	4	3	4	2

The minimum is $2 = r$ and the maximum is 4, so T_5 is a $(4, 2)$ broadcast of density $1/10$.

T_6 : $\delta_{4,3} \leq 1/8$. $L_6 = \langle (4,0), (1,1) \rangle$ of index 4 (16 vertices), invariant $c = (x-y) \bmod 4$,

$$T_6 = \{ 0@ (x, y) : x-y \equiv 0 \pmod{4} \} \cup \{ 3@ (x, y) : x-y \equiv 2 \pmod{4} \},$$

which is 2 of 16 classes, density $1/8 < 1/7$. The complete table, $u = a@(c, 0)$:

	$c = 0$	$c = 1$	$c = 2$	$c = 3$
$a = 0$	4*	4	4	4
$a = 1$	3	4	3	5
$a = 2$	3	5	3	4
$a = 3$	4	4	4*	4

The minimum over the sixteen classes is $3 = r$, so T_6 is a $(4, 3)$ broadcast of density $1/8$.

T_7 : $\delta_{4,4} \leq 5/36$. $L_7 = \langle (6,0), (0,3) \rangle$ of index 18, so a fundamental domain has 72 vertices, with class invariant $(x \bmod 6, y \bmod 3)$. Take the ten classes

$$T_7 = \{ 0@(0,0), 0@(2,0), 0@(4,1), 1@(0,2), 1@(3,0), \\ 2@(1,1), 2@(3,2), 2@(5,2), 3@(2,2), 3@(5,0) \},$$

each entry read as “all $a@(x, y)$ with $x \equiv \cdot \pmod{6}$ and $y \equiv \cdot \pmod{3}$ ”. This is 10 of 72 classes, density $10/72 = 5/36 = 0.13888\dots < 1/6$. All seventy-two values of f , with the six blocks of columns indexed by $x \bmod 6$ and the three columns of each block by $y \bmod 3$:

	$x \equiv 0$			$x \equiv 1$			$x \equiv 2$			$x \equiv 3$			$x \equiv 4$			$x \equiv 5$		
$y \equiv$	0	1	2	0	1	2	0	1	2	0	1	2	0	1	2	0	1	2
$a = 0$	4*	4	5	5	4	4	4*	4	5	4	4	4	5	4*	5	4	4	4
$a = 1$	5	5	4*	4	4	4	5	4	4	4*	5	5	4	4	4	4	4	5
$a = 2$	4	4	4	5	4*	5	4	4	4	5	4	4*	4	4	5	5	4	4*
$a = 3$	5	4	4	4	4	4	5	5	4*	4	4	5	4	4	4	4*	5	5

Every entry is at least $4 = r$ and the maximum is 5, so T_7 is a $(4, 4)$ broadcast. This proves Theorem 1.

4. THE LOWER BOUND

Lemma 4 (capping). *Let T be a $(4, r)$ broadcast and set*

$$g(u) = \sum_{v \in T, d(u,v) \leq 3} \min(4 - d(u, v), r).$$

Then $g(u) \geq r$ for all u .

Proof. If some single $v \in T$ has $4 - d(u, v) \geq r$ then its own term is $\min(4 - d(u, v), r) = r$ and we are done. Otherwise $4 - d(u, v) < r$ for every tower in the ball, so every term is unchanged by the cap and $g(u) = f(u) \geq r$. $\square$

Proof of Theorem 2. By (2) each tower v contributes $\min(4 - d, r)$ to each of the 1, 3, 5, 8 vertices at distance $d = 0, 1, 2, 3$ from it, so it contributes exactly $C(r) = \min(4, r) + 3 \min(3, r) + 5 \min(2, r) + 8 \min(1, r)$ to $\sum_u g(u)$. Summing Lemma 4 over the vertices of a window H_n and dividing by $|V(H_n)|$ gives $r \leq (|T \cap V(H_n)|/|V(H_n)|)C(r) + O(|\partial H_n|/|V(H_n)|)$, and the error term vanishes. Hence $\text{density}(T) \geq r/C(r)$ for every $(4, r)$ broadcast. Finally $C(2) = 2 + 6 + 10 + 8 = 26$, $C(3) = 3 + 9 + 10 + 8 = 30$ and $C(4) = 4 + 9 + 10 + 8 = 31$. $\square$

5. WHAT IS NOT SETTLED

- **The optimal densities.** The gaps $\delta_{4,2} \in [1/13, 1/10]$, $\delta_{4,3} \in [1/10, 1/8]$ and $\delta_{4,4} \in [4/31, 5/36]$ are open. Nothing above shows either end is attained; in particular nothing above asserts $\delta_{4,2} = 1/10$, and since $\delta_{4,r}$ is an infimum an optimal broadcast need not be periodic at all.
- **The literal form of the companion problem of [2].** That problem asks for a proof that the sets *of the paper's own three figures* dominate $H_{\infty,\infty}$ with the stated densities. Those sets appear in the source only as raster images and we did not transcribe them; the patterns above are independent objects and need not coincide with them. We therefore assert nothing whatever about the authors' figures — in particular no claim about which reception mechanism they use. The inequalities of the conjecture follow from Theorem 1, read with the reception rule (1) and the density of §2; the literal reading of that problem is untouched. The further companion question of [2] concerns the six pairs (t, r) for which $\delta_{t,r}$ is proved there, not $(4, 2)$, $(4, 3)$ or $(4, 4)$, and is not addressed here.
- **Adjacent work.** The efficient- and perfect-domination results for Archimedean lattices of Zhao, Wierman and Marge [8, 9] concern domination ratios rather than (t, r) broadcast densities, and give none of the bounds above. The (t, r) broadcast literature closest in parameters works on the infinite square grid and its higher-dimensional analogues rather than on $H_{\infty,\infty}$: Blessing, Johnson, Mauretour and

Insko [1], Drews, Harris and Randolph [3] together with their computational companion [4], Herrman and van Hintum [5], and Shlomi [6]. Those papers bound broadcast densities of grids, and none of them bounds $\delta_{4,2}$, $\delta_{4,3}$ or $\delta_{4,4}$ of the truncated square tiling. In the published version of [2] the three inequalities above are still stated as a conjecture, with the same three values, and its problem list still asks for a proof.

VERIFICATION

The exhibited objects and the hand arithmetic are re-derived from the congruences printed here by the accompanying program `verify.py`, which rebuilds the graph from the six adjacency rules of §2, performs a fresh breadth-first search in the infinite graph at every class representative, and uses exact rational arithmetic throughout.

The statement the program bears on is Theorem 1. The checks that carry it are the presentation of the graph and the shell sizes (2), and, for each of T_5, T_6, T_7 , the lattice invariance, the class count, the density and all $20 + 16 + 72$ typeset f -values, together with the comparisons $1/10 < 9/80$, $1/8 < 1/7$ and $5/36 < 1/6$; in the transcript these are the lines prefixed `graph`, `shells`, `ball-sizes`, `W5`, `W6` and `W7`. Theorem 2 is *not* verified there: the program recomputes only the integers $C(2) = 26$, $C(3) = 30$, $C(4) = 31$ and the fractions $r/C(r)$, and its own lines record that the capping step is not checked. So Theorem 2, and with it the left-hand end of the sandwich $1/13 \leq \delta_{4,2} \leq 1/10$, rests on Lemma 4 and the hand proof of §4 alone. Nor does the program establish Proposition 3: it runs the source’s own (3,1) broadcast of density $1/8$ through the (3,1) and then the (4,2) predicate, which is that one witness and not the general implication; the implication is proved above by hand.

The program’s closing line reads `VERDICT: ALL 99 CHECKS PASS`, but those 99 include checks on further patterns and comparisons this paper does not use, so that line is not by itself a certificate of the statements above: only the groups named in the previous paragraph are. The program does not read the source paper’s raster figures, which were never transcribed, so it makes no claim about them; it makes no claim about optimality either.

The tables are also short enough to check by hand, and several entries were.

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source lines: the density conjecture at 1292–1298, the companion problem on its three figures at 1417–1419, the further companion question at 1422, and the theorems on $\delta_{3,1}$, $\delta_{3,2}$ and $\delta_{3,3}$ at 666, 813 and 1077 respectively. In the published version the density conjecture is Conjecture 4.10, the problem asking for a proof for its own three figures is Problem 5, and the bounds $\delta_{3,1} \leq 1/8$ and $\delta_{3,2} \leq 1/6$ are Theorems 4.5 and 4.6.

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